A **review** of **machine learning** applications in **wildfire** science and **management**

P Jain, SCP Coogan, SG Subramanian… - Environmental …, 2020 - cdnsciencepub.com

… wide adoption of **machine learning** (ML) methods in the environmental sciences. Here, we present a scoping **review** of ML applications in **wildfire** science and **management**. Our overall …

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A systematic **review** of applications of **machine learning techniques** for **wildfire management** decision support

K Bot, JG Borges - Inventions, 2022 - mdpi.com

… **Wildfire management** decision support models are thus important for avoiding or mitigating … a **review** of recent applications of **machine learning** methods for **wildfire management** …

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Machine learning and **deep learning** for **wildfire** spread prediction: A **review**

HS Andrianarivony, MA Akhloufi - Fire, 2024 - mdpi.com

… a systematic **review** of recent ML and DL **techniques** developed for **wildfire** spread prediction, … potential for DL to play an important role in **wildfire management** and mitigation strategies. …

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Forest fire surveillance systems: A **review** of **deep learning** methods

A Saleh, MA Zulkifley, HH Harun, F Gaudreault… - Heliyon, 2024 - cell.com

… **review** aims to critically examine the existing state-of-the-art **forest fire** detection systems that are based on **deep learning** … will summarize the **review** of **forest fire** detection using DL and …

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A Comprehensive **Review** of Empirical and Dynamic **Wildfire** Simulators and **Machine Learning Techniques** used for the Prediction of **Wildfire** in Australia: H. Singh et …

H Singh, LM Ang, D Paudyal, M Acuna… - Technology, Knowledge …, 2025 - Springer

… This **review** aims to inform future research and development in … By informing future research and development, this **review** seeks to contribute to more effective **wildfire management** …

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Application of **deep learning** in **forest fire** prediction: a systematic **review**

C Mambile, S Kaijage, J Leo - IEEe Access, 2024 - ieeexplore.ieee.org

… **Deep Learning** (DL) offers a promising solution for enhancing prediction capabilities. This systematic **review** evaluated various DL **techniques** for FF … and effective DL **techniques** for FF …

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Wildfire management in Canada: **Review**, challenges and opportunities

C Tymstra, BJ Stocks, X Cai, MD Flannigan - Progress in Disaster Science, 2020 - Elsevier

… We conclude with a proposed future path for **wildfire management** in Canada. To co-exist with **wildfire**, agencies in Canada must also strengthen and adjust their **wildfire management** …

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Uncertainty and risk in **wildland fire management**: a **review**

MP Thompson, DE Calkin - Journal of environmental **management**, 2011 - Elsevier

… Beyond the unpredictability of **wildfire** behavior, uncertainty stems from inaccurate/missing … **wildfire** planning efforts. We **review** the state of **wildfire** risk assessment and **management**, …

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Spatial prediction of **wildfire** susceptibility using field **survey** GPS data and **machine learning** approaches

O Ghorbanzadeh, K Valizadeh Kamran, T Blaschke… - Fire, 2019 - mdpi.com

… of **wildfire** susceptibility. We used the capabilities of these three ML approaches using sixteen **wildfire** … The **wildfire** conditioning factors were selected according to the interpretations of …

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Review of **wildfire management techniques**—Part I: Causes, prevention, detection, suppression, and data analytics

S Jazebi, F De Leon, A Nelson - IEEE Transactions on Power …, 2019 - ieeexplore.ieee.org

… **wildfire** issues. The topics include prediction and prevention means, detection methods, monitoring and surveillance **techniques**, … and negative effects that a **wildfire** can cause to the …

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A **review** on early **wildfire** detection from unmanned aerial vehicles using **deep learning**-based computer vision algorithms

A Bouguettaya, H Zarzour, AM Taberkit, A Kechida - Signal Processing, 2022 - Elsevier

… **Wildfire** is one of the most critical natural disasters that threaten wildlands and forest … Autonomous **wildfire** early detection from UAV-based visual data using different **deep learning** …

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Role of **machine learning** algorithms in **forest fire management**: A literature **review**

M Arif, KK Alghamdi, SA Sahel… - J. Robot …, 2021 - pdfs.semanticscholar.org

… This paper aims to summarize recent trends in the **forest fire** events prediction, detection… **review** of the usage of different **machine learning** algorithms in **forest fire** or **wildfire management**…

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A **review** of recent advances in risk analysis for **wildfire management**

C Miller, AA Ager - International journal of **wildland fire**, 2013 - connectsci.au

… In this paper, we **review** recent advances in risk analysis approaches to address **wildfire management** and planning issues. We emphasise developments since the 2005 special issue …

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A **review** on **forest fire** detection **techniques**

AAA Alkhatib - International Journal of Distributed Sensor …, 2014 - journals.sagepub.com

… This work will summarise all the technologies that have been used for **forest fire** detection with exhaustive surveys of their **techniques**/methods used in this application. Methods. A lot of …

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Review of **wildfire** detection using social media

V Slavkovikj, S Verstockt, S Van Hoecke… - Fire safety journal, 2014 - Elsevier

… for **wildfire** detection and **management**, and propose possible directions of future research. … approaches in the domain of **wildfire** detection and **management** and identify and emphasize …

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A **survey** of **machine learning** algorithms based forest fires prediction and detection systems

F Abid - Fire technology, 2021 - Springer

… integration of **artificial intelligence** to automate … **survey** of the **machine learning** algorithms based forest fires prediction and detection systems. First, a brief introduction to the **forest fire** …

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A comprehensive **survey** of research towards AI-enabled unmanned aerial systems in pre-, active-, and post-**wildfire management**

SPH Boroujeni, A Razi, S Khoshdel, F Afghah… - Information …, 2024 - Elsevier

… knowledge and advance existing **techniques** in **wildfire management**. Recently, the use of … **review** and summarize a wide range of computer vision **techniques** in active-fire **management**…

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Emerging Trends in **Wildfire** Detection through the Lens of Computer Vision and **Wildfire** Emission Quantification: A Comprehensive **Survey**

K Das, J Poovvancheri, S Flesca, CR Calidonna… - IEEE …, 2026 - ieeexplore.ieee.org

… individually, and this is to suggest holistic approaches of enhancing **wildfire management**. The experience of this **review** may be a primary contributor to enhancing the early detection …

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Comparisons of diverse **machine learning** approaches for **wildfire** susceptibility mapping

K Gholamnia, T Gudiyangada Nachappa… - Symmetry, 2020 - mdpi.com

… **Wildfire** inventory data was created by integrating the … of various **machine learning** (ML) methods for **wildfire** susceptibility … (LR), which is very apt for **wildfire** susceptibility studies. The …

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A **review** of recent forest and **wildland fire management** decision support systems research

DL Martell - Current Forestry Reports, 2015 - Springer

… can be viewed from a supply chain **management** … this **review** deals with decision-making and planning concerning the suppression aspects of the forest and **wildland fire management** …

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